EXPERIMENT 12

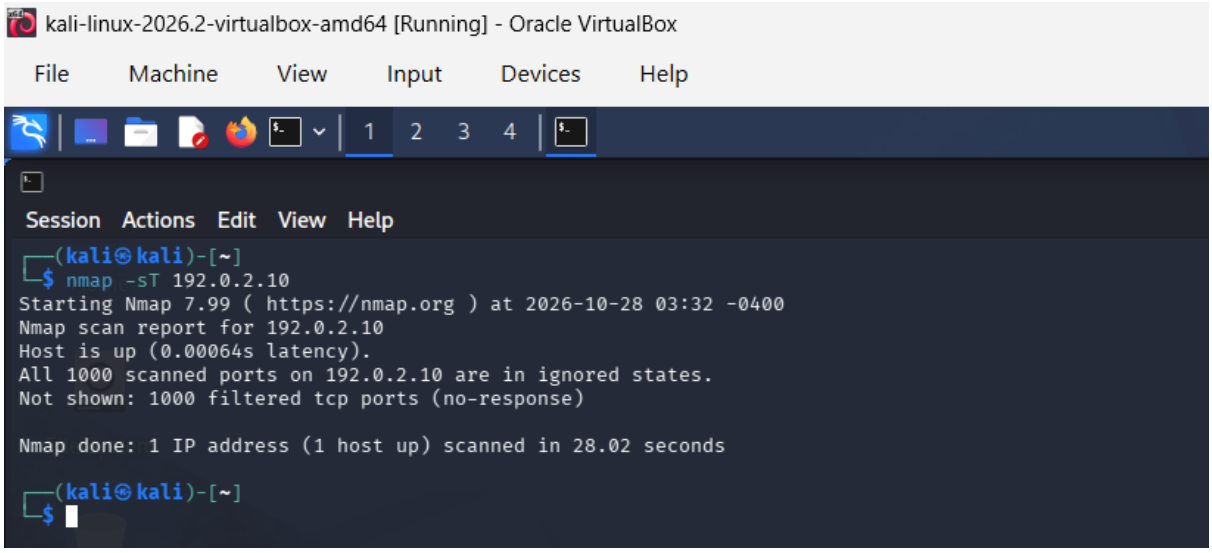
Port Scanning Tools

AIM : To identify open ports and running services on a target system using port scanning tools like Nmap for security analysis.

DESCRIPTIONS

1. nmap -sT 192.0.2.10 (TCP Connect Scan)

The nmap -sT 192.0.2.10 command performs a TCP Connect Scan to identify open ports on the target system by completing the TCP three-way handshake.



2. nmap -sS 192.0.2.10 (TCP SYN Scan)

The nmap -sS 192.0.2.10 command performs a TCP SYN Scan to identify open ports by sending SYN packets without completing the full TCP three-way handshake.

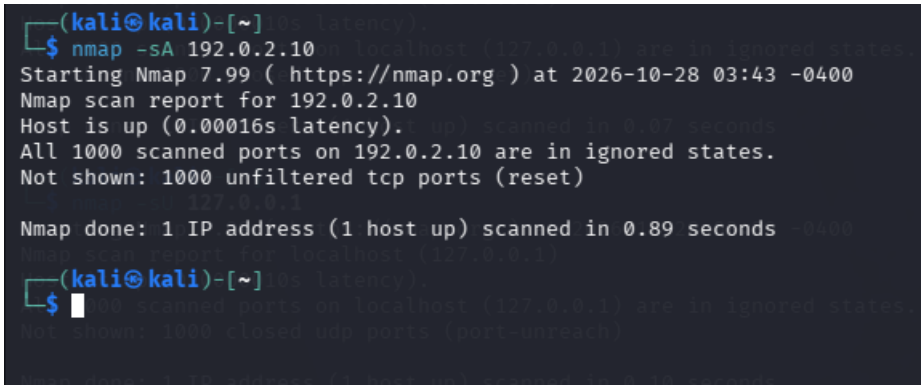


3. nmap -sU 192.0.2.10(UDP Scan)

The nmap -sU 192.0.2.10 command performs a UDP scan to identify open UDP ports and detect services running on the target system.



4. nmap -sA 192.0.2.10 (ACK Scan)



Scanning Techniques

Flag	Use	Example
-sT	TCP connect port scan	nmap -sT 192.168.1.1
-sS	TCP syn port scan	nmap -sS 192.168.1.1
–sU	UDP port scan	nmap –sU 192.168.1.1
–sA	TCP ack port scan	nmap –sA 192.168.1.1

RESULT : The ACK scan helps identify filtered and unfiltered ports on the target system and assess firewall filtering behavior.